

Northlake: The Dream Synthetic Customer

PREPARED BY Energy Hippo implementation team

Goal (north star). Turn Northlake into a complete, deterministic synthetic customer that exercises **every part of EEMSuite** — every report, every rate construct, the full AP/GL financial interface, tenant rebilling with every allocation and reconciliation method, GHG, ENERGY STAR, baselines/regression, projects, alarms, tasks/workflows, importers, and scheduled report delivery to 100+ users — so that loading Northlake into a blank EEMSuite seed proves the product end to end and produces known, repeatable expected results.

This folder is the plan for that goal. It is grounded in a full sweep of the production code at [energyhippo/EEMSuite](#), the live EEMSuite database, and the current synthetic dataset in this repo (see the dated discovery in the session that created these docs).

Current starting point (2026-09-14): use the expanded [customer packet](#) and [Student Center UI checklist](#). Build one gateway path with the existing setup account before expanding across the inventory. The broader plans below remain the coverage backlog; their product/report counts and permission details need checking when each phase is implemented.

Doc	What it is
dream-customer-overview.pdf (this document)	Goal, current state, coverage scorecard, roadmap summary, open decisions
subsystem-gap-analysis.pdf	Per-subsystem: what EEMSuite supports, what Northlake has, the gap, what to add
report-coverage-matrix.pdf	All 115 EEMSuite reports → data bucket → covered/partial/missing → phase
roadmap.pdf	Phased plan with deliverables and acceptance criteria
bill-charge-line-spec.pdf	Phase 1: how each provider's bills decompose into reconciling charge lines (→ EEMSuite BillItem)
phase1-substrate-extras.pdf	Phase 1: bill status history, digital/status points, notes & images
phase-0-1-assertions.pdf	Phase 0/1 known-value regression checklist (verify once loaded)
phase2-rate-engine-spec.pdf	Phase 2: 10 rate schedules firing all 9 determinant kinds
phase3-ap-gl-validation-spec.pdf	Phase 3: GL coding, 7 AP/GL/AR export formats, validation, accruals
phase4-tenant-rebiling-spec.pdf	Phase 4: Town Center + sub-metered Cedar Row; every allocation/WUI/reconciliation path
phase5-sustainability-analytics-spec.pdf	Phase 5: GHG, ENERGY STAR ratings, weather regression, benchmarks
phase6-operations-engagement-spec.pdf	Phase 6: projects, alarms, audit, tasks/workflows, MFR delivery to 100+, EnergyAI, HMR, importers
entry-path-and-gaps.pdf	How each thing is created via the EEM UI (not DB), and the register of UI gaps + decisions

Doc	What it is
product-gap-backlog.pdf	EEMSuite product features to build/connect to close the UI gaps (crossrepo work)

Design principles

1. **Coverage is the product.** A subsystem is "done" only when a real EEMSuite report or output renders non-empty from Northlake data with a **known expected value** we can assert against.
2. **Speak the customer's language, not EEM's.** Source data and customer-facing artifacts use real facilities/utility vocabulary (a university, a town center, a housing community). EEM-specific jargon is confined to the `data/eem/` setup translation layer. (Per the project owner: "it should fit what it does.")
3. **Deterministic & synthetic.** Same commit + seed $\Rightarrow$ same customer. Identities are fictional; only public context (weather stations, geography, utility territory shapes) is reused.
4. **New themed customers are allowed** where a subsystem needs a different shape than a campus. A mixed-use **Northlake Town Center** (retail + office + residential tower) is the natural vehicle for tenant rebilling; a larger multi-company **Northlake Portfolio** is the vehicle for data/security/permission scale and 100+ user report delivery.
5. **Model first, generate second.** Grow the synthetic **source model** to the detail each subsystem needs (charge lines, rate determinants, allocation configs, GL coding) and generate EEMSuite-ready artifacts from it, rather than hand-seeding the EEMSuite tables. This preserves determinism and traceability.
6. **Created through the UI, not the database.** The end state can be stated as "this is what must exist," but it must be reachable by **entering data through the EEM UI** (or a UI-driving system), or via a legitimate product data path (file/ODBC/handheld gateway, bill importer, external integration), or as a side effect of using the app. Direct DB seeding of customer data is off-limits. Where the UI can't create something the product supports, that is a **finding** — a missed screen or a feature to add. See [entry-path-and-gaps.pdf](#) for the per-subsystem entry paths and the gap register.

The scale of the target

- **115 reports** across 18 functional modules. **92 are MFR-deliverable** (can be scheduled/emailed). **94 are data-backed**; 21 are config/editor screens.
- The reports collapse to **~28 data-capability buckets**. Everything rests on a **substrate** every report needs: a populated hierarchy (**MetaWorld**), security (users permissioned to nodes), units/measure-types, a fiscal calendar, and meters tied to billing accounts. **Without the substrate loaded, every report is empty regardless of how rich the rest of the data is.**
- EEMSuite supports **37 measure types, 9 rate-determinant math kinds, 7 bill-allocation methods + 3 point-allocation methods, 7+ AP/GL export file formats, and 6 task types**. Northlake today touches a fraction of each.

Current onboarding packet

The generated packet (0.3) contains **3 companies, 3 sites (plus the Weather Reference site), 13 buildings, 5 providers, 49 accounts/agreements (40 external utility accounts, 6 internal cost centers, 3 PPA agreements), 322 meters (55 utility meters, 265 owned meters including 248 apartment submeters, 2 weather stations), 431 points, 139 related measurements, 7 aggregates and 14 base rate schedules**. Each of the 124 Cedar Row apartments has an electric and a water submeter. Ten buildings carry gross floor area; the Parking Structure is parking area, and Campus Grounds and Cedar Row Carport and Grounds are outdoor service areas with no floor area. Gateway mappings cover 15 profiles and 21 format cases, plus one HMR event publisher. The UI-ID capture template also includes the degree-day, baseline and aggregate outputs, for 445 point records in total. The Data Collection workbook has 26 tabs: 21 generated plus 5 retained later-phase tabs.

The first source delivery is executable: the Student Center AcquiSuite files contain a complete day, a missing-block case and its backfill, with independent totals and checksums. The workbook, register and coverage list are regenerated from the [owned source fields](#): run `python generators/northlake_packet.py`, then `python generators/update_workbook.py` to rebuild the workbook's generated tabs.

Installed UI setup and ingestion acceptance have not been recorded. The remaining gateways need source fixtures or services. Rate calculations, bills, AP/GL, permissions, tenants and the other subsystems remain later implementation and acceptance work. A database backup becomes the reference customer only after UI entry, normal product ingestion, visible checks and a restore check.

Coverage scorecard

This is the earlier broad **design** scorecard, not an installed acceptance record. Green means a modeled requirement; it does not mean a gateway or report has passed. Use the current packet's gateway coverage for the first milestone. (Detail and "what to add" in [subsystem-gap-analysis.pdf](#).)

Subsystem	EEMSuite reports/outputs	Status
Hierarchy / security / units substrate	(all reports depend on it)	● Designed, not loaded
Monthly bills + charges	Bill Processing/Tracking (~20)	● Bills modeled; no charge-line breakdown
Interval / time-series use	Use Analysis (15)	● Strong (electric, solar, CHW)
Aggregates & peak/demand	AggregateDemand, AggregatePeakLoad	● 7 aggregates defined
Rate engine	Rate Analysis (9)	● TOU/tiered/seasonal/PPA only; missing RTP, stepped, max-bill, ratchet, load-factor, %-riders, taxes, min-bill, class options, scripts, multi-currency
AP / GL / financial interface	AP Detail, AP Approval/Upload, GLAccountUDV	● Missing (no GL accounts, no UDF coding, no export formats/runs)
Tenant rebilling & allocation	Tenant Rebilling (14)	● Missing (no multi-tenant property, no allocation/WUI configs, no generated bills, no reconciliation)
Accruals (open / MPA)	OpenAccruals, accruals, mpaaccruals	● Missing
Bill validation	BillValidation, BillValidationStatus	● Missing (deferred in Stage 1)

Subsystem	EEMSuite reports/outputs	Status
GHG / emissions	GHG History/Variance/Audit (3)	● Missing (factors named in providers, not configured)
ENERGY STAR ratings	RatingComparison/History/RequestLog (3)	● Property mapping done; no rating responses/scores
Weather regression / normalization	WeatherRegression, RegressionSummary	● HDD/CDD + baseline shells; no stored regression results
Benchmarks / EUI	BenchmarkRanking, MonthlyEUI, PerformanceSnapshot	● Have floor areas; no peer set / EUI config
Project tracking	Project Detail/Summary/Finance (4)	● Missing
Alarms	AlarmDataMap	● Missing
Audit / activity logs	Audit, ActivityTracker	● Missing
Tasks / job scheduler / workflows	TaskLog, GatewayStatus, job_scheduler	● Gateways designed; no task run history / workflows
MFR saved reports & delivery	(92 deliverable reports)	● Missing (no saved reports, dashboards, deliveries, recipients)
Importers (Bill Importer, GreenButton, MyUpload/MyImport)	bill_importer, my_imports, my_exports	● Handheld/gateway files designed; no bill-importer/GreenButton/upload configs
HMR (handheld)	HMRConfiguration, hmr_export	● Upload files designed; HMR meter/route config TBD
EnergyAI	EnergyAllInsights	● Missing
Commodities / unique units	(cross-cutting)	● Good (unique units present: ton-hours, klb, ERU) — can extend

● covered/strong · ● partial · ● missing

Roadmap at a glance

Detail + acceptance criteria in [roadmap.pdf](#).

Phase specs 0–6 and their data/eem/ UI-entry plans are authored. They still require current-product review, source samples and installed acceptance. Packet 0.3 starts with company/hierarchy/provider/account/meter/point entry, then the AcquiSuite normal, replay, gap and backfill checks. Complete permission testing follows later. Expand the source model whenever a later requirement needs additional customer facts or expected results.

Phase	Theme	Unlocks
0	Foundation through the UI	Required hierarchy and measurement setup exist; first gateway acceptance can run
1	Consumption & cost core (bills w/ charge lines, intervals, units)	Use Analysis (15) + Bill Processing/Tracking core

Phase	Theme	Unlocks
2	Rate engine completeness (10 schedules → all 9 determinant kinds)	Rate Analysis (9)
3	AP / GL & financial interface (GL coding, 7 export formats, validation, accruals)	AP + accrual + validation reports; real AP interface files
4	Tenant rebilling (mixed-use property; all allocation + WUI + reconciliation)	Tenant Rebilling (14)
5	Sustainability & analytics (GHG, ENERGY STAR ratings, regression, benchmarks)	GHG (3) + ES Rating (3) + regression + benchmark
6	Operations & engagement (projects, alarms, audit, tasks, MFR delivery to 100+, EnergyAI, importers)	Remaining System Manager / Project / My Reports + scheduled delivery

Sequencing and source decisions

- 1. Sequencing (updated 2026-09-14).** Start with the minimal foundation and one AcquiSuite path using the setup account. Expand to the other gateways, relationships and aggregates, then the broader business and user scenarios.
- 2. Output approach.** Versioned source facts generate customer-facing workbooks, registers and source files. EEMSuite setup happens through the UI; data enters through supported gateways/importers. This repository does not insert customer configuration or readings directly into the EEM database.
- 3. Tenant-rebiling vehicle. ✓ Both** — a new mixed-use **Northlake Town Center** (commercial allocation) and sub-metered **Cedar Row** (residential allocation). Built in Phase 4.

Still open (decide when the phase arrives)

- **Scale targets.** How many companies/users for permission + 100+ MFR delivery testing (Phase 6). Default plan: one portfolio company with ~120 users reached via group recipients, plus a few individual/external recipients.
- **Unsupported rate constructs.** Default: document power factor, coincident peak/4-CP, CPP events, and net-metering banking as **synthetic stand-ins** (the engine can't model them natively).